* measure of position

① Quartile

[0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

Q_1 = 25%

Q_2 = 50%

Q_3 = 75%

Q_4 = 100% max

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

80%

② Percentile →

[2, 2, 3, 4, 5, 5, 5, 6, 7, 8, 8, 8, 8, 9, 9, 10, 11, 11, 12]

Percentile of 9th position

$$\text{percentile} = \frac{16}{\cancel{19+1}} \times \cancel{100}^{5}$$

$$\Rightarrow 80\%$$

Case II -

$$= \frac{18}{\cancel{19+1}} \times \cancel{100}^{5}$$

$$= \underline{\underline{90\%}}$$

③ Percentile rank -

$$= \frac{\text{Percentile}}{100} \times (n+1)$$

Case -I

$$60\% = \frac{90}{\cancel{100}_{5}} \times \cancel{19+1}$$

$$\Rightarrow 18 \text{ index}$$

$= 8$

Case-II $\quad 80\%$

$$= \frac{80}{100} \times 19 + 1$$

$$= \frac{\cancel{80}}{\cancel{5}} \quad \underline{\underline{16}}$$

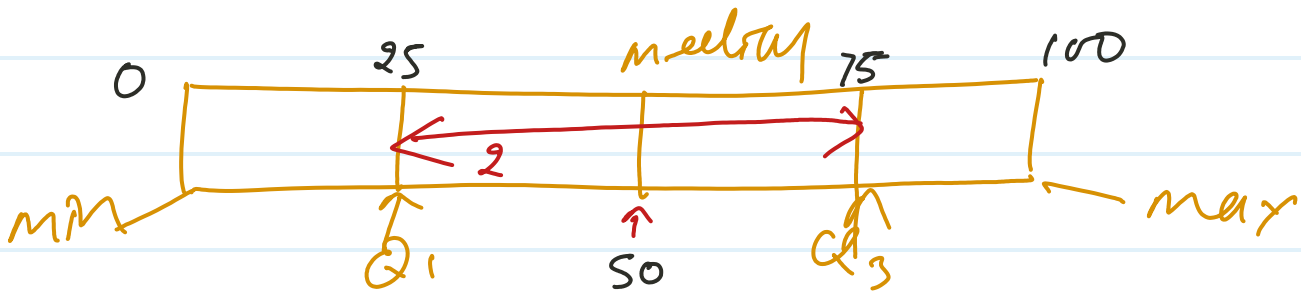
④ Interquartile Range -

$Q_1 = 25\%$

median $= 50\%$ / median

$Q_3 = 75\%$

$$IQR = Q_3 - Q_1$$



five number summary

$\Rightarrow [2, 2, 3, 4, 4, 5, 6, 7, 7, 8, 9, 9, 11, 12, 50]$

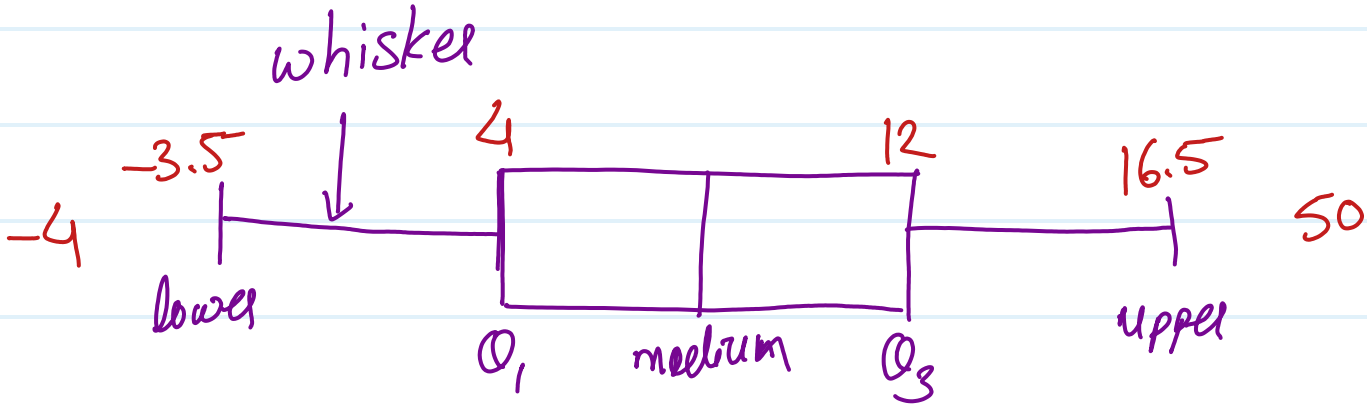
$Q_1 = \frac{25}{100} \times 15+1 \Rightarrow \frac{1}{4} \times 16^{4} \Rightarrow$ 4 index 4

$Q_3 = \frac{75}{100} \times 16 \Rightarrow \frac{3}{4} \times 16^{4} \Rightarrow$ 12 index 9

$IQR = Q_3 - Q_1 = 9 - 4 = 5$

upper fence $= Q_3 + 1.5\,IQR = 9 + 1.5 \times 5 = 16.5$

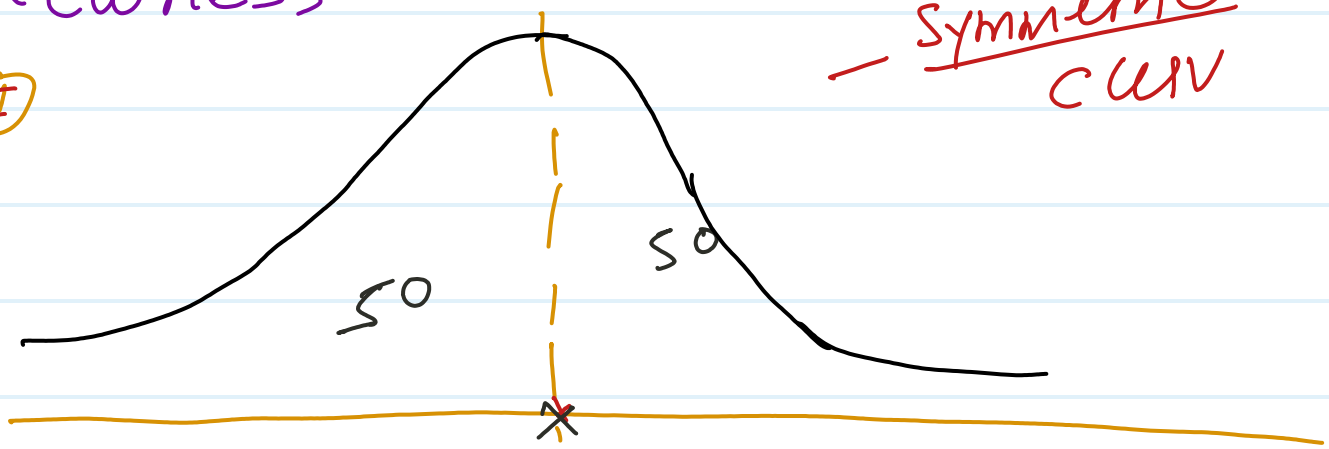
lower fence $= Q_1 - 1.5\,IQR = 4 - 1.5 \times 5 = -3.5$



✡ measure of shape

① skewness –

Ⅰ

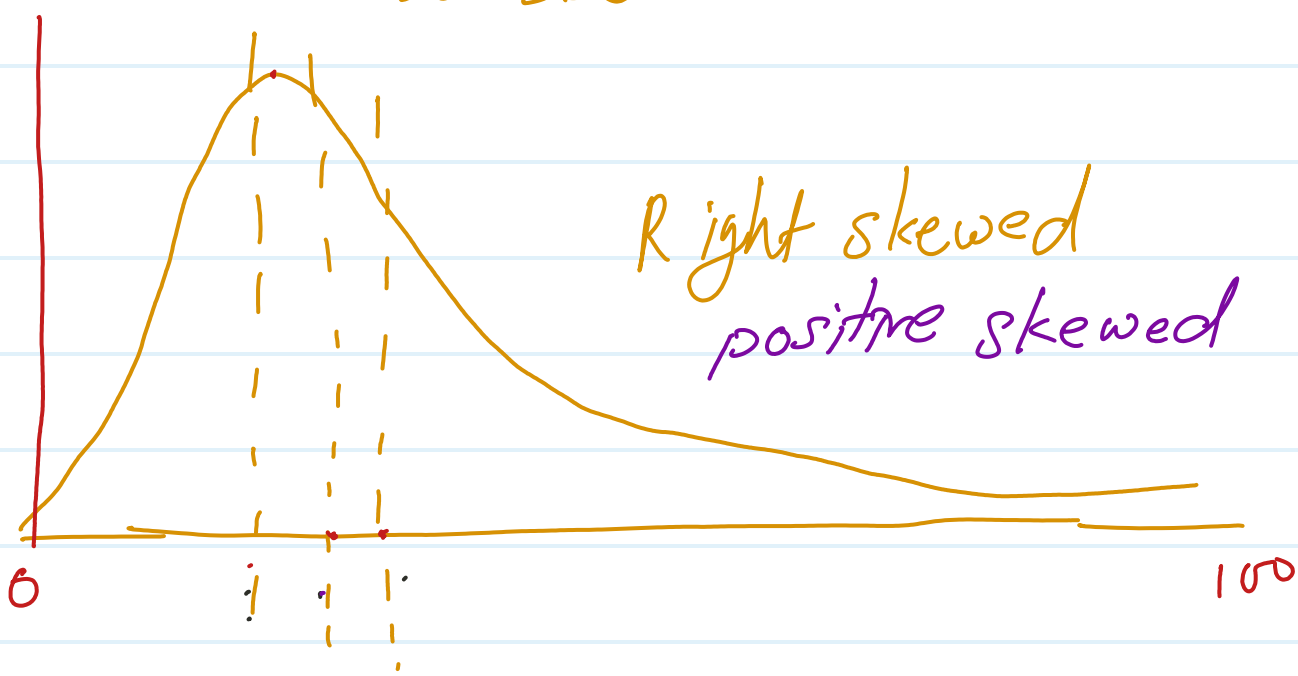


– symmetric curv

m = m = m

Zero skewed

②



Right skewed

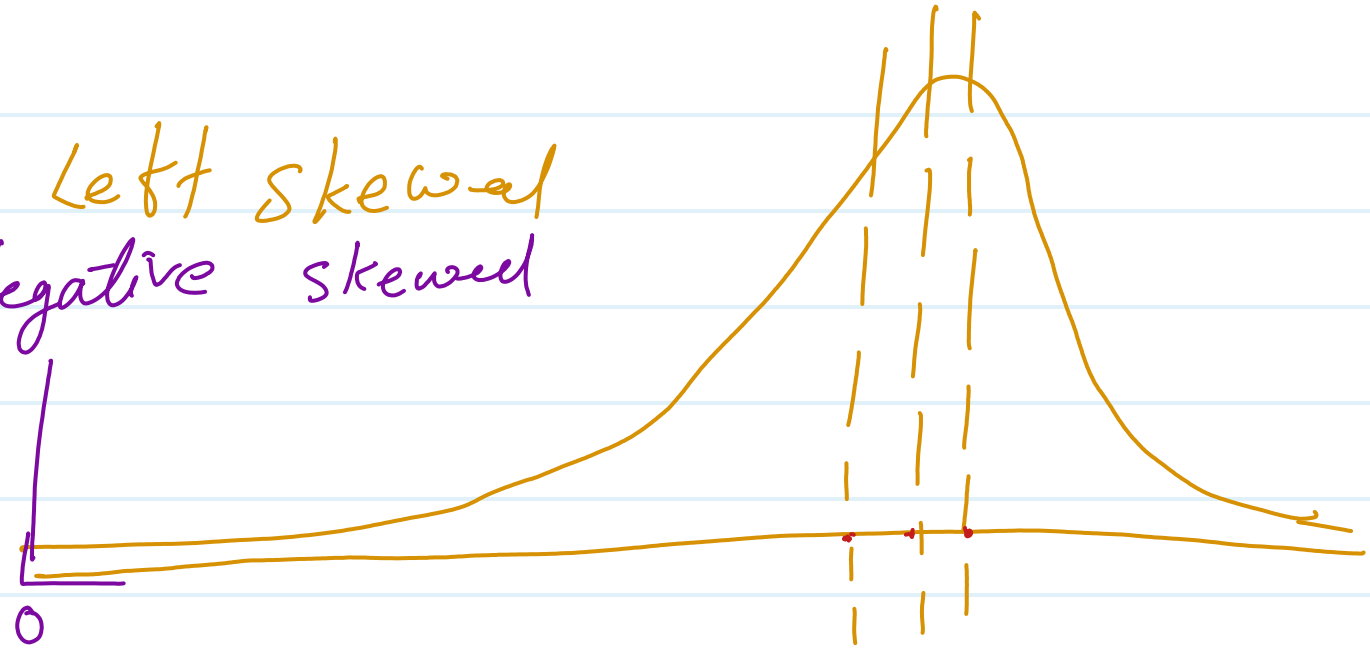
positive skewed

mean > median > mode

Asymmetric

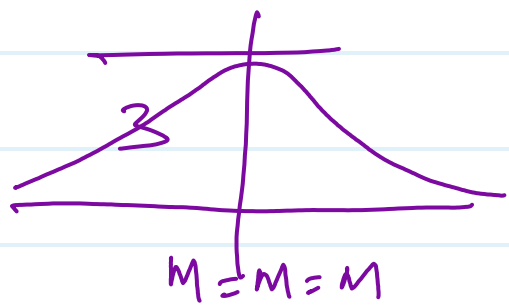
③ Left skewed

Negative skewed



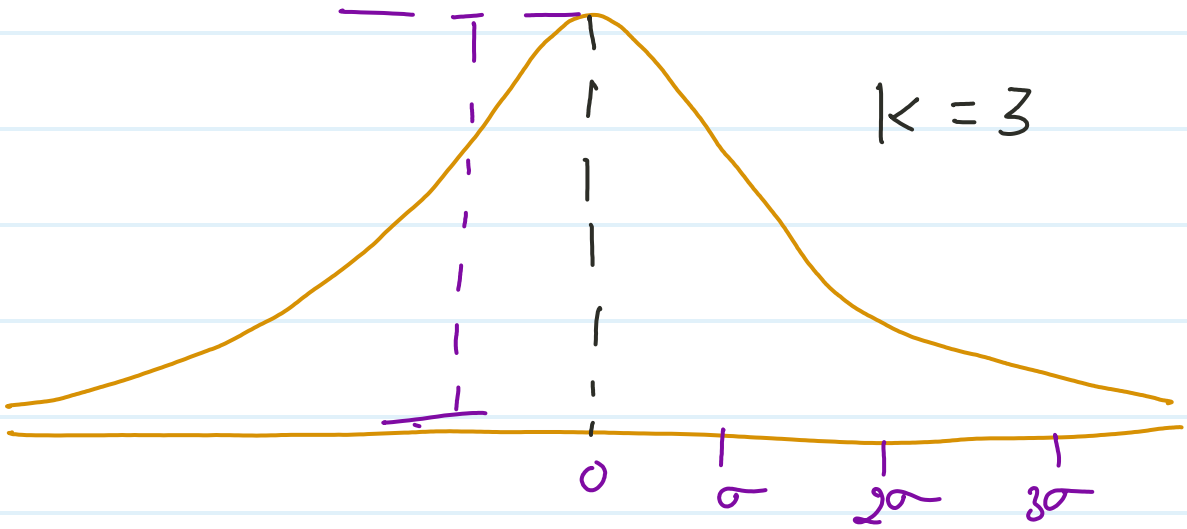
mean < median < mode

② kurtosis -



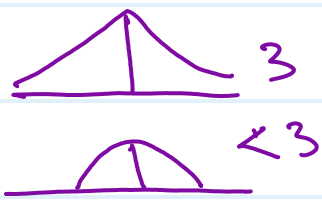
$K = 3$

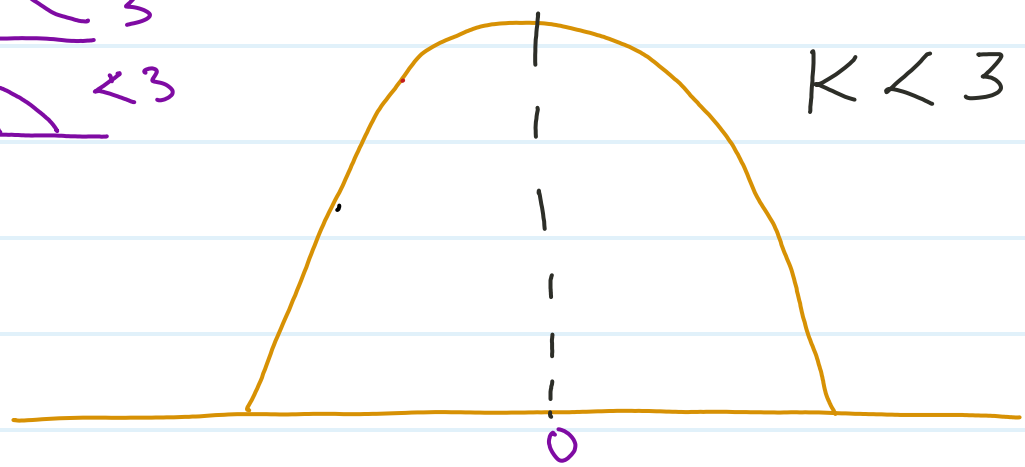
① mesokurtosis -



$K = 3$

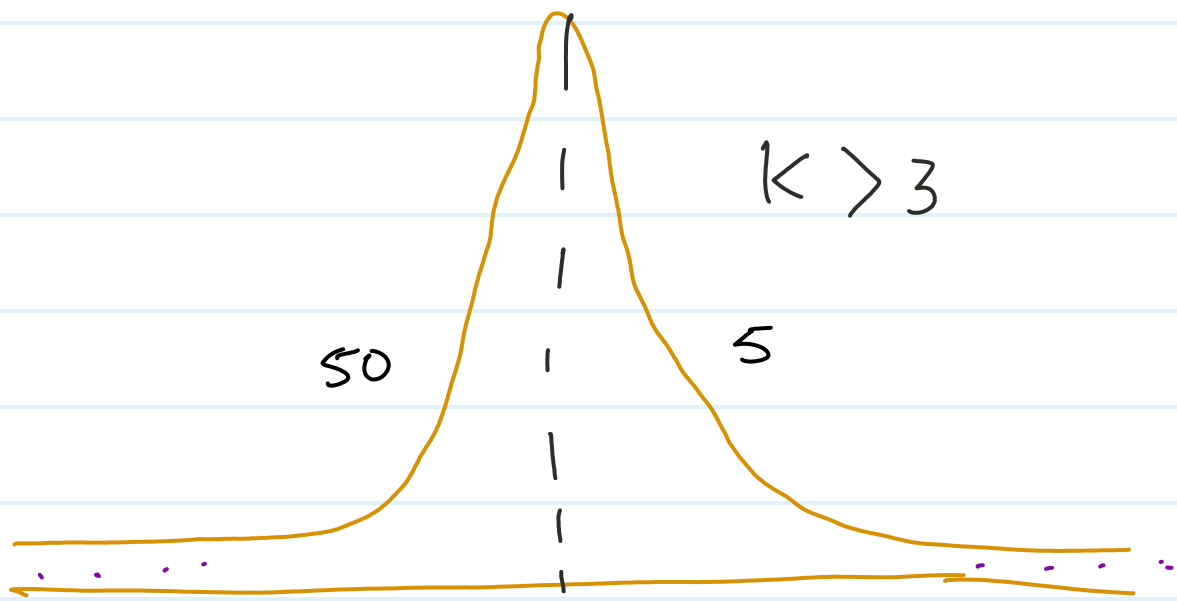
② platikurtosis –





$K < 3$

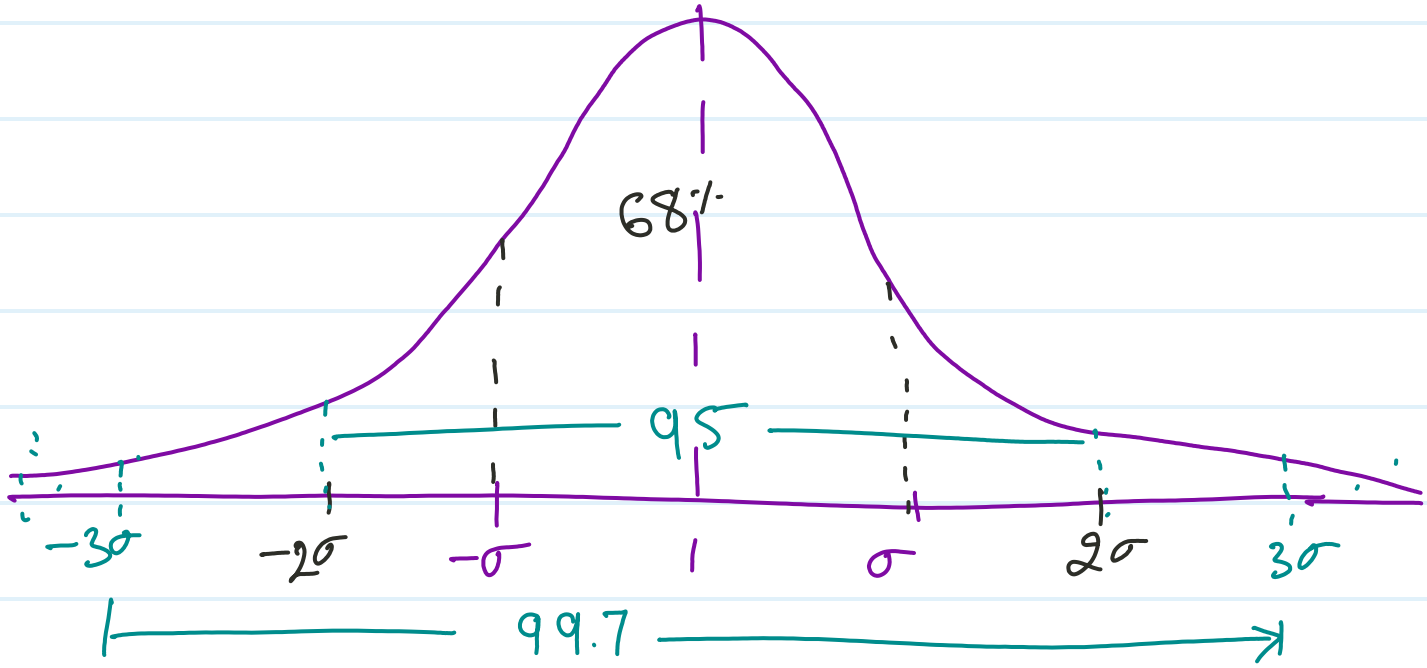
③ Lepto kurtosis –



$K > 3$

<u>Skewness</u> = –1

Emperical Rule



$\boxed{68 - 95 - 99.7}$ – Normal Dist.

Gaussian Dist.

Bell curv Dist.

③ Box and whisker plot

5 number summery

1. min
2. max
3. Q_1
4. Q_3
5. median

$$[-10, -3, -1, 2, \underset{\uparrow}{3}, 5, 6, 8, 10, 11, 12, 13, 15, \underset{\uparrow}{17}, 18, 90, 120, 150]$$

lower fence / limit
Upper fence / limit

$$Q_1 = \frac{\cancel{25}}{\cancel{100}_{4}} \times (18+1) \Rightarrow 4.75$$

$$Q_3 = \frac{\cancel{75}^{3}}{\cancel{100}_{4}} \times (18+1) \Rightarrow 14.25$$

$Q_1 = 3$
$Q_3 = 17$

$IQR = Q_3 - Q_1$

$\Rightarrow 17 - 3$

$= 14$

* lower limit $= Q_1 - 1.5\ IQR$

min

$= 3 - 1.5 \times 14$

$\Rightarrow -18$

Upper limit $= Q_3 + 1.5\ IQR$

max

$= 17 + 1.5 \times 14$

$\Rightarrow 38$

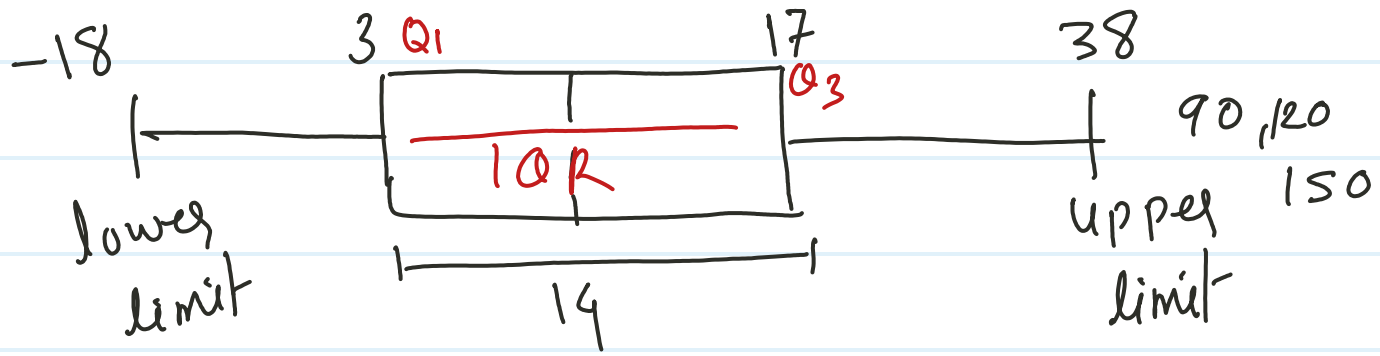
$Q_1 = 3$

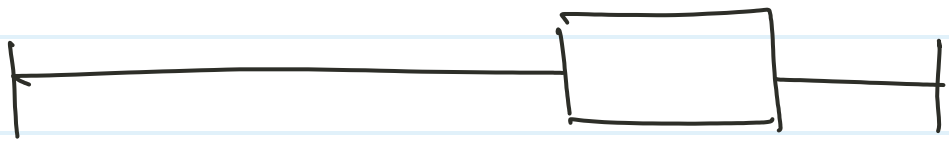
$Q_3 = 17$

lower limit $= -18$

upper limit $= 38$

$IQR = 14$





1, 2, 3, 4, 5, 6, 15 = 5.1

1, 2, 3, 4, 5, 6, 60 = 11.6

Variables

$x = 4$

Type of variable

- Quantitative
  - Descrite quan. var.
  - Continous quan. var.
- Qualitative
  - Nominal qual. var.
  - ordinal qual. var.

X_1	X_2	X_3	X_4
2	2.1	M	10th
3	3.2	F	gradate
4	4.5	T	PhD.
6	6.7	M	–
7	0.1	F	–
8	-1.1	F	–

$x = 4$

$x = x_1$

Random Variable

$X = 4$

$X \approx Y$

Type of Random variable

- Descrite Random variable
  - whole numbers
- Continous Random variable
  - Decimal,

Independent variable OR Features

Dependent variable OR Target variable

	X_1	X_2	X_3	X_4	Y
loan	✓	✓	✓	✓	—
	2		0.5		
	4		1.2		
	11		1.75		
	13				

X_1	X_2	X_3	X_4	Y
age	Cibil	Income	Job	loan
35	720	60k	Priv.	10L.
40	680	40k	Gov	20L.
25	780	75k	Pvt	30L.

Independent feature

Feature

metrics

Dependent feature

or

Target

$$\boxed{X_1 = \text{age}}$$